Chapter

5 Arithmetic Progressions

Arithmetic Progression and its nth Term

IP Important Points

Sequence: A sequence is a successions of numbers or terms formed according to some rule.

Arithmetic Progression (AP): An arithmetic progression is a list of numbers in which each term is obtained by adding a fixed number to the preceding term except the first term. This fixed number is known as common difference of the AP. Common difference of an AP can be negative, positive or zero.

General term of an AP: General term or nth term of an AP is written as a_n and $a_n = a + (n-1)d$
where a is the first term of the AP, d is the common difference and n is the number of terms.

Also $d = a_{n+1} - a_n$

If three terms a, b and c are in AP, then $b - a = c - b$ or $2b = a + c$

MCQs Multiple Choice Questions (1 Mark)

1. The next term of the A.P. : $\sqrt{6}, \sqrt{24}, \sqrt{54}$ is : [2023]

   (a) $\sqrt{60}$ (b) $\sqrt{96}$

   (c) $\sqrt{72}$ (d) $\sqrt{216}$

2. If $x + 1, 3x$ and $4x + 2$ are three consecutive terms of an A.P., then the value of x is : [2023(C)]

   (a) 2 (b) 3

   (c) 4 (d) 5

3. The first term of an AP is p and the common difference is q, then its 10th term is [2020]

   (a) $q + 9p$ (b) $p - 9q$

   (c) $p + 9q$ (d) $2p + 9q$

VSA Very Short Answer Type Questions (2 Marks)

4. Which term of the AP $-\frac{11}{2}, -3, -\frac{1}{2}, \ldots$ is $\frac{49}{2}$? [2022]

5. Find a and b so that the numbers a, 7, b, 23 are in AP. [2022]

6. Show that $(a-b)^2$, $(a^2 + b^2)$ and $(a + b)^2$ are in AP. [2020]

7. How many two digits numbers are divisible by 3? [2019]

8. Which term of the AP: 3, 15, 27, 39, . . . will be 120 more than its 21st term? [2019]

9. In an AP, if the common difference $(d) = -4$ and the seventh term (a_7) is 4, then find the first term. [2018]

10. What is the common difference of an A.P. in which $a_{21} - a_7 = 84$? [2017]

11. Find how many integers between 200 and 500 are divisible by 8. [2017]

12. Which term of the progression 20, $19\frac{1}{4}, 18\frac{1}{2}, 17\frac{3}{4}$, ... is the first negative term? [2017]

13. For what value of n, are the nth terms of two A.P.s 63, 65, 67, ... and 3, 10, 17, ... equal? [2017]

14. If seven times the 7th term of an A.P. is equal to eleven times the 11th term, then what will be its 18th term? [2017]

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15. Write the nth term of the A.P. $\frac{1}{m}, \frac{1+m}{m}, \frac{1+2m}{m}, \ldots$ [2017(C)]

16. If the nth term of the A.P. –1, 4, 9, 14, ... is 129, find the value of n. [2017(C)]

17. Find the 9th term from the end (towards the first term) of the A.P. 5, 9, 13, ..., 185. [2016]

18. For what value of k will $k + 9$, $2k - 1$ and $2k + 7$ are the consecutive terms of an A.P.? [2016]

19. The 4th term of an A.P. is zero. Prove that the 25th term of the A.P. is three times its 11th term. [2016]

20. Find the middle term of the AP 6, 13, 20, ..., 216. [2015]

21. Find the number of natural numbers between 101 and 999 which are divisible by both 2 and 5. [2014]

SA Short Answer Type Questions (3 Marks)

22. How many terms are there in an A.P. whose first and fifth terms are –14 and 2, respectively and the last term is 62. [2023]

23. Which term of the A.P. : 65, 61, 57, 53, ... is the first negative term? [2023]

24. If the pth term of an A.P. is q and qth term is p, prove that its nth term is $(p + q - n)$. [2017]

25. The digits of a positive number of three digits are in A.P. and their sum is 15. The number obtained by reversing the digits is 594 less than the original number. Find the number. [2016]

26. Divide 56 in four parts in A.P. such that the ratio of the product of their extremes (1st and 4th) to the product of means (2nd and 3rd) is 5 : 6. [2016]

27. The pth, qth and rth terms of an A.P. are a, b and c respectively. Show that $a(q - r) + b(r - p) + c(p - q) = 0$ [2016]

28. The sum of the 5th and the 9th terms of an AP is 30. If its 25th term is three times its 8th term, find the AP. [2014]

LA Long Answer Type Questions (5 Marks)

29. The sum of four consecutive numbers in AP is 32 and the ratio of the product of the first and last terms to the product of two middle terms is 7 : 15. Find the numbers. [2020]

30. Which term of the Arithmetic Progression – 7, –12, –17, –22, ... will be –82. Is –100 any term of the A.P.? Give reason for your answer. [2019]

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